

Handling Class Imbalance and Final Model Selection

Week 5, Day 3

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Abstract

This session addresses one of the most pervasive challenges in healthcare machine learning: class imbalance. We will transition from baseline evaluations to advanced techniques designed to improve the detection of the minority class (e.g., patients with heart disease).

- **The Problem:** Understanding the "Accuracy Paradox" and the clinical cost of False Negatives.
- **The Solution:** Introducing SMOTE (Synthetic Minority Over-sampling Technique) and the critical rules for preventing data leakage.
- **Retraining:** Architecting imbalanced pipelines and re-evaluating models using rigorous cross-validation.
- **Final Selection:** Comparing baseline and SMOTE metrics to make a clinically sound, deployment-ready model recommendation.

Note: Explainable AI (XAI/SHAP) is reserved for a future session.

Session Objectives

- Analyze the baseline confusion matrix to understand the inherent Precision-Recall trade-off.
- Explain why Accuracy is a dangerous and misleading metric in imbalanced medical datasets.
- Master the conceptual mechanics of SMOTE and the strict rules for preventing data leakage.
- Architect and execute retraining pipelines that safely incorporate synthetic oversampling.
- Compare baseline and SMOTE performance to make a final, clinically justified model selection.

- ➊ **The Class Imbalance Problem in Healthcare** (Accuracy Paradox, Precision vs. Recall)
- ➋ **Introduction to SMOTE** (Concept, Interpolation, and the Golden Rule of Leakage Prevention)
- ➌ **Retraining Models with SMOTE** (ImbPipeline architecture, RandomizedSearchCV, untouched test set)
- ➍ **Performance Comparison & Final Model Selection** (Metric shifts, trade-off analysis, clinical recommendation)

Part 1: The Class Imbalance Problem in Healthcare

"Why standard metrics fail in medical diagnostics."

The Reality of Class Imbalance in Healthcare

The Natural State of Medical Data

In clinical datasets, the "positive" class (e.g., presence of a rare disease, adverse drug reaction, or heart disease) is almost always the minority.

- **The Algorithmic Bias:** Machine learning algorithms are inherently designed to minimize overall error.
- **The Consequence:** When faced with imbalance, the algorithm learns that the safest way to minimize error is to simply predict the majority class every time.
- **The Result:** A model that is mathematically "accurate" but clinically useless, as it fails to identify the very patients who need intervention.

Reviewing the Baseline Confusion Matrix

The Baseline Reality

Our initial models, trained on imbalanced data, likely produced a confusion matrix heavily skewed toward True Negatives and False Negatives.

- **True Negatives (TN):** High. The model easily identifies healthy patients.
- **False Negatives (FN):** Unacceptably high. The model misses a significant portion of diseased patients.
- **True Positives (TP):** Low. The model struggles to confidently flag the minority class.

This baseline reveals the urgent need for intervention to shift the model's decision boundary.

The Precision-Recall Trade-off

The Seesaw Effect

Precision and Recall are intrinsically linked and often move in opposite directions when adjusting a model's classification threshold.

- **Increasing Recall:** Casting a wider net to catch more true positives inevitably catches more false positives, which *lowers* Precision.
- **Increasing Precision:** Making the model more conservative to ensure flagged cases are truly positive means missing some actual positives, which *lowers* Recall.

The Challenge

The goal is not to maximize both simultaneously (which is mathematically impossible), but to find the optimal operating point that aligns with clinical risk tolerance.

The "Accuracy Paradox"

Definition

The Accuracy Paradox occurs when a model achieves a high accuracy score, but this score is entirely driven by its performance on the majority class, masking terrible performance on the minority class.

- **Example:** In a dataset with 95% healthy patients and 5% diseased patients, a "dumb" model that always predicts "Healthy" achieves 95% accuracy.
- **The Illusion:** Reporting 95% accuracy sounds impressive to stakeholders, but the model has a 0% Recall for the disease.
- **The Takeaway:** In imbalanced medical datasets, Accuracy is a dangerous, misleading metric that must be discarded in favor of Recall, F1-Score, and PR-AUC.

The Clinical Imperative: Maximizing Recall

The Asymmetric Cost of Errors

In healthcare, the cost of a False Negative (missing a disease) vastly outweighs the cost of a False Positive (unnecessary testing).

- **False Negative Cost:** Patient is sent home untreated, potentially leading to severe morbidity or mortality.
- **False Positive Cost:** Patient undergoes additional, non-invasive screening. This incurs financial cost and temporary anxiety, but preserves life.

Our Primary Objective

We must engineer the model to maximize Recall (Sensitivity), ensuring we capture as many true positive cases as possible, while maintaining an acceptable, non-catastrophic level of Precision.

Part 2: Introduction to SMOTE

"Synthetic Minority Over-sampling Technique."

What is SMOTE?

Synthetic Minority Over-sampling Technique (SMOTE)

SMOTE is an advanced data augmentation technique designed to balance class distributions by generating *synthetic* (artificial) examples of the minority class, rather than simply duplicating existing ones.

- **Origin:** Proposed by Chawla et al. (2002) specifically to address the limitations of random oversampling.
- **Goal:** To expand the decision region of the minority class, giving the machine learning algorithm more "examples" to learn the underlying patterns of the rare disease.

How SMOTE Works: The Interpolation Concept

The Algorithmic Mechanism

SMOTE operates in the feature space using k-Nearest Neighbors (k-NN).

- ❶ **Select:** Choose a random instance from the minority class.
- ❷ **Identify:** Find its k nearest neighbors *within the same minority class*.
- ❸ **Interpolate:** Randomly select one of these neighbors. Create a new, synthetic instance at a random point along the line segment connecting the original instance and the chosen neighbor in the feature space.

Result: Plausible, novel data points that preserve the statistical properties of the minority class without exact duplication.

SMOTE vs. Random Oversampling

Random Oversampling	SMOTE
Duplicates existing minority instances exactly.	Generates <i>new</i> , synthetic instances via interpolation.
High risk of severe overfitting (model memorizes exact duplicates).	Reduces overfitting risk by introducing slight, realistic variations.
Does not add new information to the feature space.	Expands the decision boundary of the minority class.

Key Takeaway

SMOTE provides a more robust and generalized representation of the minority class, leading to better real-world performance than naive duplication.

The Golden Rule of SMOTE: Preventing Data Leakage

The Most Critical Concept in This Session

SMOTE must **ONLY** be applied to the training data (`X_train`, `y_train`).

- **The Forbidden Action:** Applying SMOTE to the entire dataset *before* the train-test split.
- **The Consequence:** Severe, catastrophic data leakage.
- **Why?** If SMOTE is applied globally, synthetic minority instances are generated using information from what will eventually become the test set. The model is then trained on data that is mathematically derived from the test set.

Why Pre-Split SMOTE Causes Severe Leakage

The Illusion of Perfection

If you apply SMOTE before splitting, your cross-validation and test scores will look miraculously high (e.g., 98% Recall).

- **The Reality:** This performance is entirely artificial. The model has effectively "memorized" the test set through the synthetic interpolation process.
- **The Deployment Failure:** When deployed on truly unseen, real-world patient data, the model's performance will collapse to baseline levels or worse, because it never learned to generalize; it only learned to recognize leaked patterns.

Rule: The test set must remain a pristine, untouched representation of the real-world class imbalance.

The Correct Workflow: SMOTE on Training Data Only

The Safe, Leak-Proof Sequence

1. Split the raw data into Train and Test sets.
2. Apply preprocessing (imputation, scaling) *only* to the Train set.
3. Apply SMOTE *only* to the preprocessed Train set to balance it.
4. Train the model on this balanced, preprocessed Train set.
5. Evaluate on the *original, imbalanced, untouched* Test set.

Why This Works

The model learns from a balanced representation of the world, but is judged on its ability to navigate the messy, imbalanced reality of the real world.

Part 3: Retraining Models with SMOTE

"Architecting safe pipelines and re-evaluating performance."

Architecting the Imbalanced Pipeline (ImbPipeline)

The Concept of ImbPipeline

To enforce the "Golden Rule" automatically and prevent human error, we use specialized pipeline structures (e.g., `imblearn.pipeline.Pipeline`).

- **Standard Pipeline:** Preprocessing → Classifier.
- **ImbPipeline:** Preprocessing (ColumnTransformer) → Resampling (SMOTE) → Classifier.

The Magic of ImbPipeline

During cross-validation, the ImbPipeline ensures that SMOTE is fitted and applied *independently within each training fold*, completely isolating the validation fold from any synthetic data generation.

The Order of Operations in ImbPipeline

Why Preprocessing Must Precede SMOTE

The sequence within the pipeline is strictly enforced for mathematical validity.

- ❶ **Step 1: ColumnTransformer.** Handles missing values and scales features. SMOTE relies on distance calculations (k-NN); if features are not scaled, distance metrics will be dominated by variables with larger magnitudes, creating poor synthetic samples.
- ❷ **Step 2: SMOTE.** Generates synthetic samples in the properly scaled, clean feature space.
- ❸ **Step 3: Classifier.** Trains on the balanced, preprocessed data.

Reversing Steps 1 and 2 is a common, critical error that degrades SMOTE's effectiveness.

Re-running RandomizedSearchCV with SMOTE

The Tuning Process Revisited

We wrap our new ImbPipeline into the RandomizedSearchCV framework, exactly as we did in Session 2, but with a crucial difference in the underlying data flow.

- **The Search Space:** Remains the same (e.g., tuning tree depth, learning rates).
- **The Evaluation Metric:** Remains `roc_auc` (or `average_precision`), as these are robust to the class imbalance we are actively trying to fix.
- **The Execution:** For every hyperparameter combination, the pipeline safely preprocesses the fold, applies SMOTE to the training portion of the fold, trains the model, and scores it on the untouched validation portion of the fold.

The Role of the Untouched Hold-Out Test Set

The Ultimate Arbiter of Truth

After identifying the best SMOTE-tuned pipeline, we must evaluate it on the `X_test` and `y_test` sets.

- **Crucial Detail:** We do **NOT** apply SMOTE to the test set.
- **Why?** The test set must reflect the true, natural prevalence of the disease in the target population.
- **The Test:** Can the model, which was trained on a balanced, synthetic view of the world, successfully identify the rare positive cases in a realistically imbalanced environment?

Evaluating SMOTE-Trained Models

What We Expect to See

Compared to the baseline, the SMOTE-trained models should exhibit a distinct shift in their confusion matrix.

- **True Positives (TP):** Should increase significantly. The model is now "aware" of the minority class.
- **False Negatives (FN):** Should decrease significantly. This is our primary clinical goal.
- **False Positives (FP):** Will likely increase. This is the expected, acceptable trade-off for catching more true positives.

The success of SMOTE is measured by whether the gain in TP outweighs the cost of the increase in FP.

Part 4: Performance Comparison & Final Model Selection

"Making the clinically sound deployment decision."

Baseline vs. SMOTE: The Metric Shift

The Comparative Analysis

We now place the baseline metrics and the SMOTE metrics side-by-side to quantify the impact of our intervention.

- **Accuracy:** May actually *decrease* slightly. Do not panic. This is normal and expected, as the model is no longer lazily predicting the majority class.
- **Recall:** Should show a marked, substantial improvement. This validates that SMOTE successfully taught the model to recognize the minority class.
- **ROC-AUC:** Should improve, indicating better overall separability between the classes.

The Primary Indicators of Success

In this clinical context, these two metrics tell the most important story.

- **Recall (Sensitivity):** Did we successfully reduce the number of missed diagnoses (False Negatives)? A jump from 60% to 85% Recall is a massive clinical victory.
- **F1-Score:** Because Recall and Precision are inversely related, the F1-Score provides a single, balanced metric. An improved F1-Score indicates that the model gained Recall without completely sacrificing Precision to the point of uselessness.

The Importance of PR-AUC

Precision-Recall Area Under Curve (PR-AUC)

While ROC-AUC is excellent, PR-AUC is the superior metric for evaluating models on imbalanced datasets.

- **The Focus:** PR-AUC ignores True Negatives entirely. It focuses exclusively on the model's performance regarding the positive (minority) class.
- **The Interpretation:** A high PR-AUC means the model maintains high Precision even as it pushes Recall to its limits.
- **The Verdict:** If the SMOTE model shows a significantly higher PR-AUC than the baseline, it is definitively the better model for this specific clinical task.

Analyzing the Trade-offs: The Precision Cost

The Inevitable Question

"Did SMOTE cause a severe drop in Precision?"

- **Scenario A (Ideal):** Recall increases significantly, Precision drops slightly or remains stable. *Verdict: SMOTE was highly successful.* **Scenario B (Acceptable):** Recall increases dramatically, Precision drops moderately, but F1-Score and PR-AUC improve. *Verdict: SMOTE is clinically justified. The cost of extra tests is worth the lives saved.* **Scenario C (Failure):** Recall increases marginally, but Precision collapses, leading to an overwhelming number of False Positives that would paralyze the clinical workflow. *Verdict: SMOTE parameters need tuning, or a different algorithm is required.*

Did SMOTE Work? Interpreting the Results

Holistic Interpretation

Success is not defined by a single number, but by the alignment of the metrics with the clinical objective.

- If the SMOTE-trained model successfully shifts the confusion matrix to capture more True Positives, and the PR-AUC confirms this is a robust pattern rather than random noise, then SMOTE has achieved its goal.
- The model is now "calibrated" to be appropriately sensitive to the minority class, reflecting the true clinical priorities of the project.

Beyond the Metrics: The Deployment Checklist

Selecting the final model requires balancing statistical performance with practical clinical constraints.

- ➊ **Primary Metric:** Highest Recall, provided Precision does not fall below a clinically acceptable minimum threshold.
- ➋ **Secondary Metric:** Highest PR-AUC and F1-Score, indicating a robust balance.
- ➌ **Stability:** Low variance in cross-validation scores (the model is reliable, not a lucky fluke).
- ➍ **Computational Efficiency:** Reasonable training and inference times for the intended deployment environment (e.g., real-time EHR integration vs. overnight batch processing).

Candidate 1: Random Forest (The Stable Performer)

Profile

An ensemble of decorrelated decision trees trained on balanced data via SMOTE.

- **Strengths:** Highly robust to overfitting, even when SMOTE introduces synthetic noise. Provides stable, well-calibrated probability estimates.
- **Clinical Fit:** Excellent choice if the priority is a reliable, consistent model that is less prone to extreme False Positive spikes. It offers a very safe, dependable baseline for clinical deployment.
- **Trade-off:** May have a slightly lower ceiling for maximum Recall compared to more aggressive boosting algorithms.

Candidate 2: XGBoost (The High-Recall Contender)

Profile

A gradient boosting ensemble trained on SMOTE-balanced data.

- **Strengths:** Often achieves the absolute highest Recall and PR-AUC by aggressively correcting errors from previous trees. Highly effective at capturing complex, non-linear interactions in clinical data.
- **Clinical Fit:** The ideal choice if the absolute minimization of False Negatives is the paramount, non-negotiable objective, and the clinical workflow can absorb a higher rate of False Positives.
- **Trade-off:** Requires careful tuning to prevent overfitting to the synthetic SMOTE samples, and probability outputs may require calibration (e.g., Platt scaling) for true risk estimation.

Making the Final Clinical Recommendation

The Decision Framework

Based on our comparative analysis, we select the model that best satisfies the project's specific clinical goals.

- **If the goal is "Do no harm, catch everyone":** We recommend **XGBoost**, accepting the higher False Positive rate as the necessary cost of maximizing sensitivity and ensuring no diseased patient slips through the cracks.
- **If the goal is "Balanced, reliable screening":** We recommend **Random Forest**, offering a superior, more stable trade-off between Recall and Precision, preventing the clinical team from being overwhelmed by false alarms.

The final choice is a clinical decision, informed by data, not just a mathematical one.

Summary & Best Practices

- ❶ **Beware the Accuracy Paradox:** Never rely on Accuracy for imbalanced medical datasets.
- ❷ **SMOTE is Powerful, but Dangerous:** It effectively balances data, but **must** only be applied within the training folds to prevent catastrophic data leakage.
- ❸ **Use ImbPipeline:** Automate the safe sequence of Preprocessing → SMOTE → Classification.
- ❹ **Evaluate Holistically:** Judge success by the improvement in Recall, F1-Score, and PR-AUC on the *untouched* test set.
- ❺ **Align with Clinical Goals:** The final model selection must balance statistical performance with the real-world tolerance for False Positives vs. False Negatives.

Questions?

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Next Session Preview: Explainable AI (XAI) and SHAP values for model interpretability.